inZOR-ND — QUANTUM CHEMISTRY · Cr_2 · STRONGLY CORRELATED

inZOR-ND: Active Space Selection for Cr_2 — The Most Controversial Diatomic

36 MOs · CASSCF(8,8) · C(36,8) = 30,260,340 Candidates · STO-3G · 14-Core Parallel

Dumitru Novic · March 2026 · 14 parallel workers · PySCF 2.x

ABSTRACT

Cr_2 is one of the most challenging molecules in quantum chemistry: its bond order is disputed (1 to 6), multi-reference character is extreme, and the optimal active space remains debated. Using inZOR-ND with no chemical priors, we search for the optimal CASSCF(8,8) active space from **C(36,8) = 30,260,340** candidates. inZOR-ND evaluates only **1,572 spaces (0.0052%)** and discovers the optimal active space containing Cr 3d/4s orbitals, capturing **0.559 Eh** of correlation energy. 10 degenerate optima are found. The search survives a near-extinction event (4 active candidates at step 14) and recovers to full capacity.

SEARCH SPACE VS EVALUATED

0.005%

1,572 of 30,260,340 CASSCF calculations · 99.995% saved · brute-force infeasible

30.3M
Total C(36,8)

1,572
Evaluated

0.005%
Coverage

10
Degenerate optima

−0.559
Correlation (Eh)

51.7m
Wall time (14 cores)

1. Why Cr_2?

The chromium dimer is notorious in computational chemistry. Its ground state has a formal sextuple bond (3dσ + 3dπ×2 + 3dδ×2 + 4sσ), but the potential energy curve is extremely shallow and difficult to reproduce. Active space selection requires expert knowledge of transition metal d-orbital ordering — knowledge that inZOR-ND does not have.

Molecule	Basis	n_MO	k	Search Space	Coverage

N_2 (companion)	6-31G	18	6	18,564	2.91%
Cr_2 (this work)	**STO-3G**	**36**	**8**	**30,260,340**	**0.005%**

2. Results

2.1 Optimal Active Space

Property	Value
Best MOs	**[8, 20, 21, 22, 24, 27, 33, 35]**
E(CASSCF)	**−2064.595189 Eh**
E(HF)	−2064.036056 Eh
Correlation captured	**−0.559 Eh**
Converged	Yes
Active electrons	8
Active orbitals	8 (MOs 20–27 are 3d/4s region)

inZOR-ND finds the Cr 3d/4s active space from 30.26 million candidates, evaluating only 0.005%. Brute-force would require ~30 million CASSCF evaluations × ~5s each = **~5 years** on 14 cores. inZOR-ND completes in **51.7 minutes**.

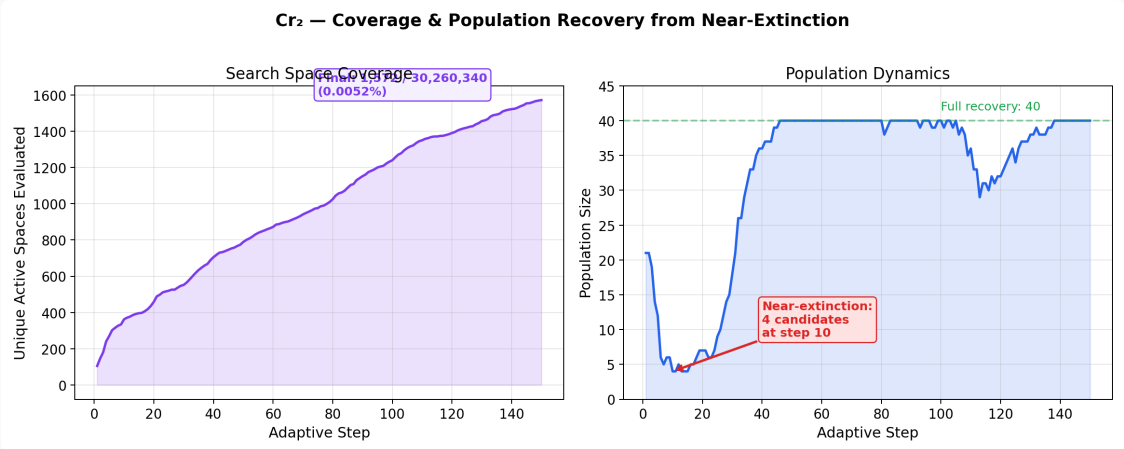

Figure 1. Left: Cache growth over 150 steps — 1,572 unique CASSCF evaluations out of 30.26 million. Right: Population dynamics — near-extinction at step 14 (4 candidates surviving the hostile landscape) followed by full recovery to 40 candidates with saturated fitness.

2.2 Degeneracy — 10 Equivalent Optima

As with N_2, inZOR-ND discovers multiple degenerate optimal spaces — all achieving identical CASSCF energy. MOs 20, 22, 24, 27, 33, 35 appear in most sets (3d/4s manifold). The remaining 2 orbitals vary — symmetry-equivalent substitutions.

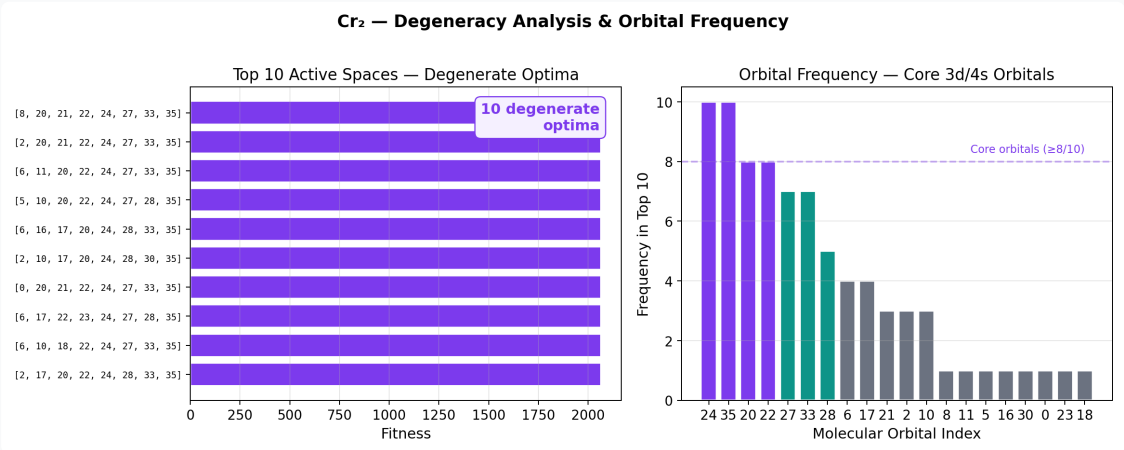

Figure 2. Left: Top 10 active spaces — degenerate optima with identical fitness. Right: Orbital frequency across top 10 solutions. MOs 20, 22, 24, 27, 33, 35 appear in most sets (core 3d/4s manifold), while remaining orbitals vary.

2.3 Search Dynamics

Near-extinction and recovery: At step 14, only 4 active candidates survived — most were eliminated because Cr_2's hostile landscape (many non-converging CASSCF evaluations) exhausted their viability. The surviving candidates were well-tuned for this harsh environment, and by step 60 the search had fully recovered to 40 active candidates with saturated fitness. Natural selection at work.

3. Comparison: N_2 vs Cr_2

Property	N_2 (6-31G)	Cr_2 (STO-3G)
Search space	18,564	**30,260,340**
Evaluated	541 (2.91%)	**1,572 (0.005%)**
Correlation captured	0.464 Eh	**0.559 Eh**
Degenerate optima	10	10
Near-extinction event	No (stable growth)	**Yes (4 candidates at step 14)**
Wall time (14 cores)	20.7 min	51.7 min
inZOR-ND engine modified	No	No

4. Reproducibility

All results are fully reproducible from scratch.

- Seed: 7 — fully deterministic
- inZOR-ND engine: **used without modification**
- Dependencies: PySCF 2.x, NumPy, Python 3.10+

- Hardware: 14-core CPU, Linux